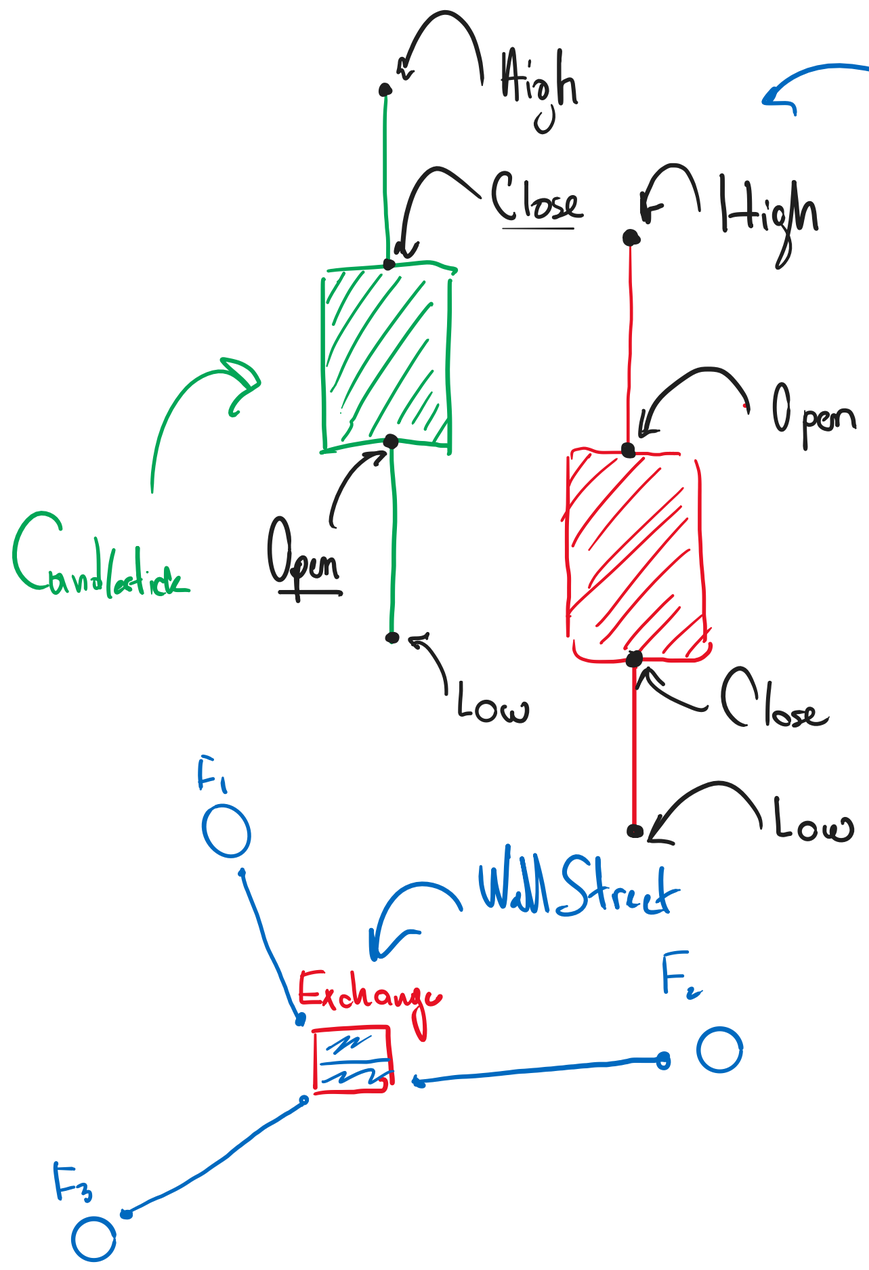

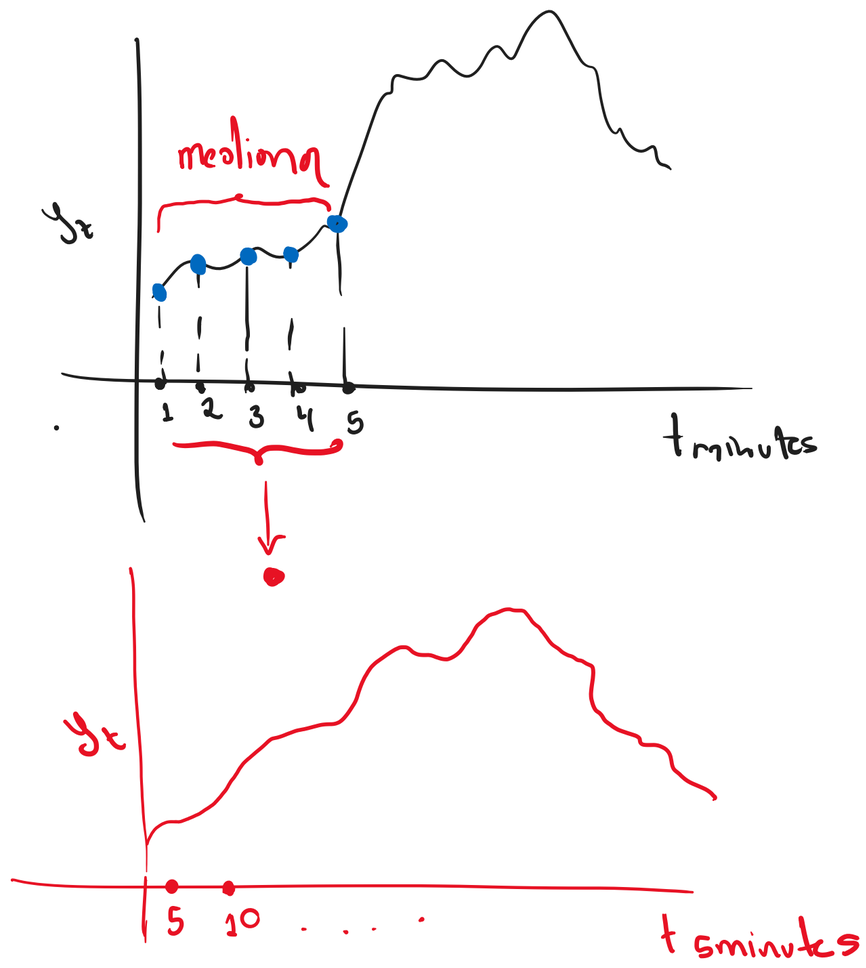

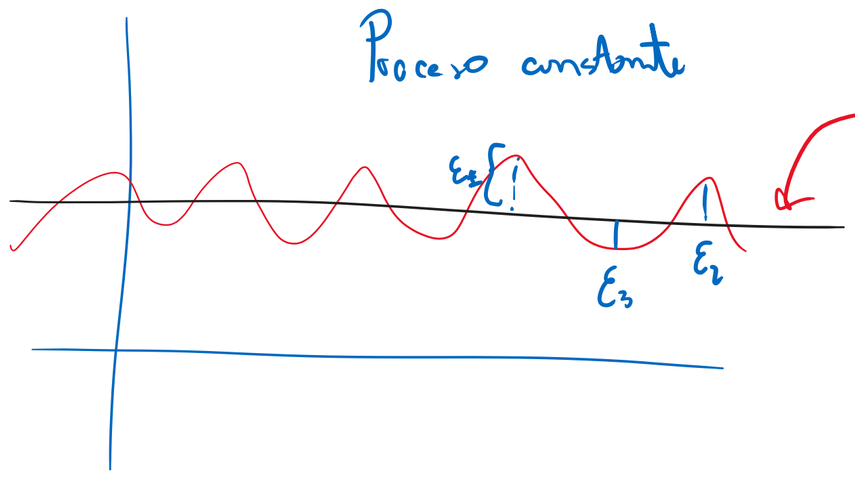

μ β_0

$$y_t = f(t,\beta) + \varepsilon_t$$

$$y_t = \beta_0 + \varepsilon_t$$

$$\hat{\beta}_0 \approx \beta_0$$

$$\hat{\beta}_0 = (1-\theta)\cdot\sum_{t=0}^{T-1}\theta^t y_{T-t}$$

$$SS_E = \sum_{t=1}^{T}(y_t-\mu)^2 \sim SS_E^* = \sum_{t=0}^{T-1}\theta^t\cdot(y_{T-t}-\beta_0)^2 \; ; \quad |\theta|<1 \text{ Geométrica}$$

$$2\beta_0 = 0$$

$$\frac{\partial}{\partial\beta_0}SS_E^* = \frac{\partial}{\partial\beta_0}\left(\sum_{t=0}^{T-1}\theta^t\cdot(y_{T-t}-\beta_0)^2\right) = \sum_{t=0}^{T-1}\theta^t\cdot 2(y_{T-t}-\beta_0)(-1)) = 0 \Rightarrow$$

$$\sum_{t=0}^{T-1}\theta^t(y_{T-t}-\beta_0) = 0 \Leftrightarrow \sum_{t=0}^{T-1}\theta^t y_{T-t} = \beta_0\sum_{t=0}^{T-1}\theta^t \Leftrightarrow$$

$$\hat{\beta}_0 = \frac{1}{\sum_{t=0}^{T-1}\theta^t}\cdot\sum_{t=0}^{T-1}\theta^t y_{T-t} = \frac{1}{\frac{1-\theta^T}{1-\theta}}\cdot\sum_{t=0}^{T-1}\theta^t y_{T-t} = \frac{1-\theta}{1-\theta^T}\cdot\sum_{t=0}^{T-1}\theta^t\cdot y_{T-t}$$

$$\frac{1-\theta^T}{1-\theta} = \sum_{t=0}^{T-1}\theta^t \quad |\theta|<1$$

$$\xrightarrow{T\to\infty} = (1-\theta)\cdot\sum_{t=0}^{T-1}\theta^t\cdot y_{T-t}$$

$$\tilde{y}_T = \hat{\beta}_0 + \varepsilon_t$$

$$= (1-\theta)\sum_{t=0}^{T-1}\theta^t\cdot y_{T-t} + \varepsilon_t$$